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**A Hybrid Approach Based on Bacterial Foraging Optimization and
the Cultural Algorithm for Data Clustering**

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Dedication

I dedicate this thesis, with boundless gratitude and appreciation and with the greatest pride and thankfulness, to:

My dear father and mother, for all the loving efforts they have made throughout the different stages of my life. Their benevolent spirit and uplifting prayers have accompanied me on my path, and through their kindness they taught me how to live.

The wise and learned professors who assisted me on the path toward knowledge and understanding.

Those who guided me in the pursuit of knowledge.

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Praise and gratitude are due to Almighty God, who granted me success and accompanied me on my path so that I could bring this thesis to completion. I sincerely thank my distinguished and thoughtful supervisor, Dr. Hassan Rashidi, who has always treated the author with kindness and consideration.

Abstract

The enormous volume of information and the need to use information efficiently have led to the emergence of a branch of science known as data mining. Pattern extraction, pattern evaluation, and obtaining useful knowledge from data related to a particular domain are among the most important characteristics of data mining. Various techniques exist for extracting useful patterns, one of which is clustering. Cluster analysis is an important technique in the growing field known as exploratory data mining and has been applied in various engineering and scientific disciplines, including biology, psychology, medicine, marketing, computer science, and satellite mapping. The use of metaheuristic algorithms, such as particle swarm optimization, genetic algorithms, and others, to overcome the weaknesses of traditional algorithms such as K-Means has received considerable attention in recent years. In this research, data clustering is performed using the Bacterial Foraging Optimization Algorithm together with the Cultural Algorithm. The Bacterial Foraging Optimization Algorithm is inspired by the foraging behavior of groups of bacteria. Alongside this algorithm, the Cultural Algorithm defines and explains the historical experience of the bacteria and establishes norms that assist the bacteria in finding optimal solutions. Experiments conducted in this study on standard datasets show that applying the bacterial algorithm with the assistance of the Cultural Algorithm to the data-clustering problem produces acceptable results and improves the evaluation criteria.

Keywords: data clustering; Bacterial Foraging Optimization Algorithm; Cultural Algorithm

Table of Contents

Chapter 1: Introduction	2
1-1. Problem Statement	3
1-2. Innovation	3
1-3. Assumptions	4
1-4. Thesis Structure	4
Chapter 2: Concepts and General Principles	5
2-1. Introduction	6
2-2. Emergence of Data Mining	8
2-3. Data Mining Process	10
2-3-1. Problem Definition and Hypothesis Formulation	11
2-3-2. Data Collection	11
2-3-3. Data Preparation	11
2-3-4. Data Mining	12
2-3-5. Model Interpretation and Result Extraction	16
2-4. Large Datasets	17
Chapter 3: Research Background	21
3-1. Introduction	22
3-2. The Need for Clustering	25
3-3. The Clustering Process	27
3-4. User Dilemmas	31
3-4-1. Data Representation	31
3-4-2. Purpose of Grouping	32
3-4-3. Number of Clusters	32

Table of Contents (Continued)

3-4-4. Cluster Validity	33
3-5. Clustering Techniques	36
3-5-1. Hierarchical Clustering Algorithm	36
3-5-2. Partitional Clustering	38
3-5-3. Nearest-Neighbor Clustering	41
3-6. Cluster Ensembles	41
3-6-1. Diversity	41
3-6-2. Consensus Function	42
3-7. Summary and Conclusions	43
Chapter 4: Proposed Data-Clustering Method	44
4-1. Introduction	45
4-2. Bacterial Foraging Optimization Algorithm	46
4-2-2. Operation of the Bacterial Foraging Method	47
4-3. Cultural Algorithm	49
4-3-1. Belief Space	50
4-3-2. Knowledge Components	51
4-3-3. Communication Protocols	51
4-4. Data Clustering by Combining the Cultural Algorithm and Bacterial Algorithm	52
4-5. Summary and Conclusions	55
Chapter 5: Experiments and Evaluation of Results	56
5-1. Introduction	57
5-2. Evaluation Criteria	57
5-2-1. Intra-Cluster Distance	57
5-2-2. Error Rate	58

Table of Contents (Continued)

5-3. Experiments	58
5-4. Summary and Conclusions	63
Chapter 6: Conclusions and Future Work	64
6-1. Summary of Previous Work	65
6-2. Conclusions	65
6-3. Future Work	66
References	67

List of Tables

Table 3-1. Different Methods for Generating Diversity	42
Table 3-2. Functions Presented in Different Articles	43
Table 5-1. Parameter Values and Variables of the Proposed Algorithm	59
Table 5-2. Summary of Dataset Characteristics	59
Table 5-3. Comparison of Results Based on the Intra-Cluster Distance Criterion	60
Table 5-4. Comparison of Results Based on Standard Deviation	61
Table 5-5. Comparison of Results Based on the Error-Rate Criterion	62

List of Figures

Figure 2-1. Block Diagram of the Parameter-Estimation Process	9
Figure 2-2. Cluster Diversity	14
Figure 2-3. Iterative Data-Mining Process	17
Figure 2-4. Tabular Representation of a Dataset	20
Figure 3-1. Data Clustering	23
Figure 3-2. Clustering Procedure	25
Figure 3-3. Finding Subclasses in Data Using Clustering	26
Figure 3-4. Merging Data in Hierarchical Clustering	28
Figure 3-5. A Typical Sequence of the First Three Steps	28
Figure 3-6. A Classification of Clustering Techniques	29
Figure 3-7. Single-Linkage Clustering	29
Figure 3-8. Importance of Good Representation	33
Figure 3-9. Effect of Different Feature Weights on the Results	34
Figure 3-10. Automatic Selection of the Number of Clusters	35
Figure 3-11. Points Within Three Clusters	37
Figure 3-12. Structure Obtained Using the Single-Link Algorithm	37
Figure 4-1. Bacterial Movement	46
Figure 4-2. Pseudocode of the Bacterial Algorithm	48
Figure 4-3. Operation of the Cultural Algorithm	49
Figure 5-1. Comparison of the Proposed Methods with Other Methods Based on Intra-Cluster Distance	61
Figure 5-2. Comparison of the Proposed Methods with Other Methods Based on Error Rate	69

Chapter 1

Introduction

1-1. Problem Statement

Data clustering is one of the data-exploration methods and is generally described as belonging to the descriptive spectrum of data mining. By discovering patterns and relationships within large datasets, it seeks to obtain an understanding of the system.

Clustering is an unsupervised classification method in which the classes are not defined in advance. One objective of data clustering is to discover natural groupings within a collection of patterns, points, or objects. Cluster analysis is defined as a statistical classification technique for determining whether the members of a population fall into different groups on the basis of quantitative comparisons of several characteristics.

One important issue in clustering algorithms is the existence of a powerful solution for clustering data. To date, no algorithm has been presented that can extract clusters of every possible shape from a dataset. Furthermore, selecting an appropriate clustering method is not straightforward and requires knowledge of the data and of the characteristics of the algorithm being used.

Clustering is suitable for problems involving data extraction, data mining, decision-making, and machine learning, including information retrieval and image segmentation.

In general, clustering methods are divided into two categories: crisp (hard) and fuzzy (soft). These categories include hierarchical and non-hierarchical methods [1]. In addition to well-known algorithms such as K-Means, powerful metaheuristic (evolutionary) algorithms are capable of solving clustering problems effectively and have produced acceptable results in this field.

1-2. Innovation

Considering the existing problems in clustering, some of which were mentioned in the previous section, this thesis attempts, as far as possible, to present solutions to these problems, including:

- Presenting methods based on swarm intelligence for clustering and for optimal search within the very large data space.
- Improving data-clustering performance through the use of a hybrid algorithm.
- Increasing the speed with which data clusters are identified.

- Using the Cultural Optimization Algorithm, which provides high accuracy in solving large and discrete problems, particularly the clustering problem. It not only increases the accuracy of information exploration but also increases the speed of that exploration.
- Using an appropriate cost function to measure the accuracy of the clusters generated during clustering.

1-3. Assumptions

- A two-dimensional matrix is used to represent the data.
- Euclidean distance is used to determine similarity between data points.
- The output of this work consists of the data identified within different clusters.
- The results of this method are compared with those of other existing evolutionary algorithms.

1-4. Thesis Structure

This thesis consists of six chapters. The first chapter presents the concepts and the general framework of the thesis. The second chapter provides an introduction to data mining, and the third chapter reviews previous methods for data clustering. The fourth chapter presents the proposed method, which is based on bacterial foraging and the Cultural Optimization Algorithm, for improving clustering performance. The results obtained from the method are then presented and compared with previous methods. The conclusions are stated in the sixth chapter. Finally, the glossary used in the thesis and the implementation of the algorithm described in this thesis are included as appendices.

Chapter 2

Concepts and General Principles

2-1. Introduction

Data mining is the process of discovering patterns and hidden information in data. There is a distinction between data in a database and data in a data warehouse. In a database, data are presented in a structured form, whereas in a data warehouse they may or may not be presented in a structured form.

In data mining, it is therefore necessary to make the data compatible with the defined processing operation. First, attention must be given to cleaning the data so that they become suitable for subsequent processing. Data cleaning may be regarded as noise removal, noise reduction, or feature removal.

The data-cleaning process can be performed using tools such as ETL, through commercially available tools, or by applying various suitable techniques. One important consideration in data mining is whether the data being handled are static or dynamic. Working with a static dataset is considerably easier than working with diverse and dynamic datasets.

For static data, all data required for the intended analysis are available before processing and generally do not vary with time. Dynamic data, however, refer to a continuously changing volume of information for which neither the final state of the data nor all the data required for processing or analysis are yet available.

Data mining requires an algorithm or method for analyzing the data of interest. The data may be sequence data¹, sequential data², time-series data, temporal signals, audio signals, or video streams.

In practice, prediction³ and description⁴ are the two main objectives of data mining. Prediction uses variables or fields within a dataset to forecast future values of other variables. Description, by contrast, focuses on finding patterns that describe the data in a form interpretable by humans. Therefore, the field of data-mining activity can be placed into the following two groups:

- 1. Predictive data mining:** given a dataset, it produces a model of the system.
- 2. Descriptive data mining:** using the available dataset, it produces new and meaningful information.

Footnotes: ¹ Sequence data. ² Sequential data. ³ Prediction. ⁴ Description.

In predictive data mining, the objective is to produce a model in the form of executable code that can be used for prediction, classification, estimation, or other similar tasks. In descriptive data mining, however, patterns and relationships within large datasets are discovered in order to understand the system.

The relative importance of predictive and descriptive data mining may vary considerably for particular applications. Prediction and description are achieved through data-mining techniques. Some of the most common data-mining techniques are listed below [1].

- 1. Classification⁵.** A learning model is extracted that assigns data samples to one of several predefined classes.
- 2. Regression⁶.** A predictive learning model is produced that maps a data sample to a real-valued prediction variable.
- 3. Clustering⁷.** An operation that searches for a finite set of groups or clusters in order to describe the data.
- 4. Summarization⁸.** Another descriptive operation that includes methods for finding a concise description of a dataset or a subset of it.
- 5. Dependency modeling⁹.** A model is produced that identifies the most significant dependencies among variables, or among the values of a feature, within a dataset or part of a dataset.
- 6. Change and deviation detection¹⁰.** This task discovers the most significant changes within a dataset.

Data mining is one of the expanding branches of computer science. One of its greatest strengths is the wide range of methods and techniques that can be applied to a very large number of problems.

Footnotes: ⁵ Classification. ⁶ Regression. ⁷ Clustering. ⁸ Summarization. ⁹ Dependency modeling. ¹⁰ Change and Deviation Detection.

In business and economics, data mining can be used to discover customers' purchasing trends, support investment planning, and identify unnecessary costs in accounting systems. It can also improve marketing competitiveness, resulting in greater attention to customers.

Many legal and investigative centers responsible for identifying illegal activity, fraud, and criminal conduct have successfully employed data mining. It has also been used in intelligent centers that retain information related to national security.

2-2. Emergence of Data Mining

A review of the different definitions of data mining proposed by various authors shows that there is no general agreement on a single definition of data mining, or even on its constituent components. Is data mining a form of statistics combined with learning theory, or is it a newly evolved concept?

In our view, most data-mining problems and their solutions are rooted in classical data analysis. Data mining has roots in several sciences, the most important of which are statistics and machine learning. Statistics is rooted in mathematics; therefore, before testing a problem experimentally, we tend to establish it theoretically. Machine learning, by contrast, is rooted in computer practice and emphasizes experimentation and implementation. Together, these two disciplines direct us toward applied problems for which suitable preparation for testing and execution exists.

If the most important distinction between statistics and machine learning in data mining is their relative emphasis on mathematics, modeling, and algorithms, another distinction is that modern statistics is derived almost entirely from the concept of the model. Statistics emphasizes modeling, whereas machine learning emphasizes algorithms. The term learning implicitly includes the concept of an algorithmic process.

System identification generally consists of two top-down stages:

- 1. Structure determination.** At this stage, prior knowledge of the target system is required in order to select the most appropriate model. The model is usually represented as a parametric function $y = f(u, t)$, where y is the model output, u is the input vector, and t is the parameter vector.

Determining the function f depends on the problem and is based on the designer's experience and understanding of the target system.

- 2. Parameter determination.** In the second stage, once the model structure has been identified, optimization techniques are used to determine the parameter vector so that the final model $y^* = f(u, t^*)$ can appropriately describe the system.

In general, system identification is not a single-pass process. Both structure determination and parameter determination must be performed repeatedly until a satisfactory model is obtained. This iterative process is shown in Figure 2-1. The general steps in each iteration are as follows:

1. Specify a mathematical model and its parameters, $y^* = f(u, t)$, representing the identified system.
2. Perform parameter estimation to select the most appropriate parameters for the available dataset so that the difference $y - y^*$ is minimized.
3. Conduct a system-validation test to evaluate whether the resulting model responds correctly to previously unseen observations. This is often called testing, validation, or system verification.
4. Terminate the process when the validation results are satisfactory.

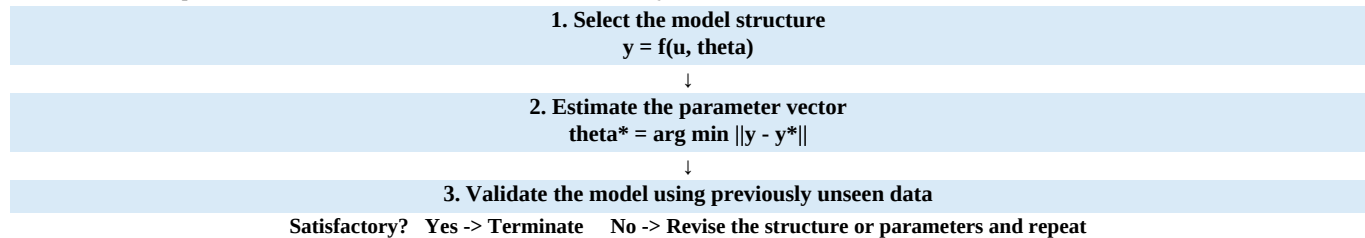


Figure 2-1. Iterative System-Identification and Parameter-Estimation Process

If no prior knowledge of the target system is available, determining the structure will be very difficult.

In that case, it must be selected through trial and error. Although the structures of most engineering systems and industrial processes are known, the structures of most target systems for which data-mining techniques are used are unknown, or are so complex that obtaining an appropriate mathematical model is impossible. Consequently, new parameter-estimation techniques have been developed and are now regarded as part of the spectrum of data-mining techniques.

We can now distinguish between the terms model and pattern in data mining. A model is a relatively broad structure involving a large number of samples, whereas a pattern is a local structure involving only some samples or a small region of the data space. It should be noted that the meaning of pattern in data mining differs from its meaning in pattern recognition. In pattern recognition, a pattern is a measurement vector that describes a particular object in a multidimensional space. In data mining, by contrast, a pattern is a local model.

2-3. Data Mining Process

This section discusses the characteristics of data mining and how data mining operates.

Constructing a data-mining model can be compared with any industrial production process. First, raw material is required, namely data. The data are then passed through a mechanical process, namely an algorithm, and finally a product, namely a data-mining model, is produced. The difference between industrial production and data mining is that mathematical concepts are used instead of mechanical equipment to produce the product.

The extraction of different models and valid patterns from a dataset is called the data-mining process. The word process is particularly important. Even in some professional environments, data mining is mistakenly believed to be simply a computer-based tool for selecting and applying a solution to a problem. There are several reasons why this view is incorrect. First, data mining is actually a collection of tools, each different from the others and each seeking the best match with the problem. The second reason concerns matching the problem with a particular technique. In general, the data-mining process includes the following stages.

2-3-1. Problem Definition and Hypothesis Formulation

Most studies on data modeling have been conducted in specific application domains. Therefore, experience and an appropriate background are necessary in order to arrive at a clear and meaningful problem statement. Unfortunately, many applied studies have focused on data-mining techniques rather than on clarifying the problem statement.

At this stage, the designer usually prepares a general form of the problem as a set of requirements and variables, expressed as an initial hypothesis. Several hypotheses may be generated for a single problem. In the first step, the problem expert's experience is combined with the data-mining model. In practice, this means that close interaction is established between the data-mining expert and the domain expert. In successful data-mining problems, this cooperation does not end after the initial stage but continues throughout the data-mining process.

2-3-2. Data Collection

This stage focuses on how data are generated and collected. In general, there are two separate approaches. The first is used when the data-generation process is supervised by the domain expert or problem designer. This approach is known as a designed experiment. The second is used when the domain expert cannot influence the data-generation process. It is known as the observation-based approach. For example, random data generation in most data-mining problems is an observation-based method.

After the data have been collected, the distribution of the samples is usually completely unknown. It is important that the data used to build the model and the data used to test the model be obtained through similar sampling. Otherwise, the constructed model may not produce successful results.

2-3-3. Data Preparation

In an observation-based approach, data are usually collected from existing databases. In general, preprocessing includes at least two broad stages:

- 1. Identification and removal of outlying samples.** Outlying samples are data items whose unusual values are inconsistent with most observations. These values usually result from errors in measurement, coding, or data recording, although they may sometimes be natural but uncommon values.

Such unusual samples can have serious effects on the production of the final model. Two strategies are available for dealing with these samples:

- a. Identify and remove outlying samples as part of the data-preprocessing phase.
- b. Develop robust modeling methods that are insensitive to outlying samples.

2. Data scaling, encoding, and feature selection. Data preprocessing includes several stages such as scaling, encoding, and the selection of features and variables. For example, in a data-mining technique, a feature with the range [0, 1] and another feature with the range [-100, 1000] do not have equal weights, and their values may affect the final data-mining result differently.

Encoding methods also reduce the dimensionality of the data for the modeling process by providing fewer and more useful features. These two data-preprocessing methods are only examples from the broad range of preprocessing methods used in the data-mining process.

The data-preprocessing stage should not be regarded as separate from the other stages of data mining. In every iteration of the data-mining process, all operational stages can work together to define a new and improved dataset for subsequent iterations. An appropriate preprocessing method, using prior knowledge in the form of scaling and feature encoding, provides a desirable representation for the data-mining technique.

In general, data preprocessing can be divided into two stages: data preparation and dimensionality reduction. For further study of different data-preprocessing methods, refer to [2] and [3].

2-3-4. Data Mining

The main task of this stage is to select and implement an appropriate data-mining technique. This is not a simple process. In practice, implementation is based on several models, and selecting the best model is another task. Some data-mining techniques used to construct models are presented below [4].

2-3-4-1. Classification and Prediction

Classification is the process of finding a set of models or functions that describe and distinguish data according to their meaningful classes and that can predict the class of samples whose class labels are unknown. The resulting models are based on analysis of a training dataset consisting of samples with known class labels.

A model derived from classification may be represented in different forms, including classification rules, decision trees, mathematical formulas, or neural networks. A decision tree is a flowchart-like tree structure in which each internal node tests the value of a feature, each branch represents an outcome of the test, and each leaf assigns a class label to the samples. Decision trees can readily be converted into classification rules. A neural network is a set of linear threshold units that can be trained to distinguish samples with different class labels.

Classification can be used to predict the class of data. In many applications, however, unknown numerical values are predicted rather than class labels. When the predicted values are numeric, the task is more precisely called prediction. Prediction may refer both to predicting a data value and to predicting a data class, although the term is usually restricted to value prediction and is therefore distinguished from classification.

Examples of classification and prediction include:

- Detecting fraud in credit-card transactions.
- Diagnosing the presence of a particular disease.
- Predicting stock value three months into the future.
- Predicting the reduction in traffic fatalities in the following year after speed restrictions are imposed.

2-3-4-2. Clustering

The objective of data clustering, also known as cluster analysis, is to discover natural groupings within a collection of patterns, points, or objects. An example of clustering is shown in Figure 2-2.

The objective is to develop an automatic algorithm that discovers the natural groupings shown in Figure 2-2(b) within the unlabeled data shown in Figure 2-2(a).

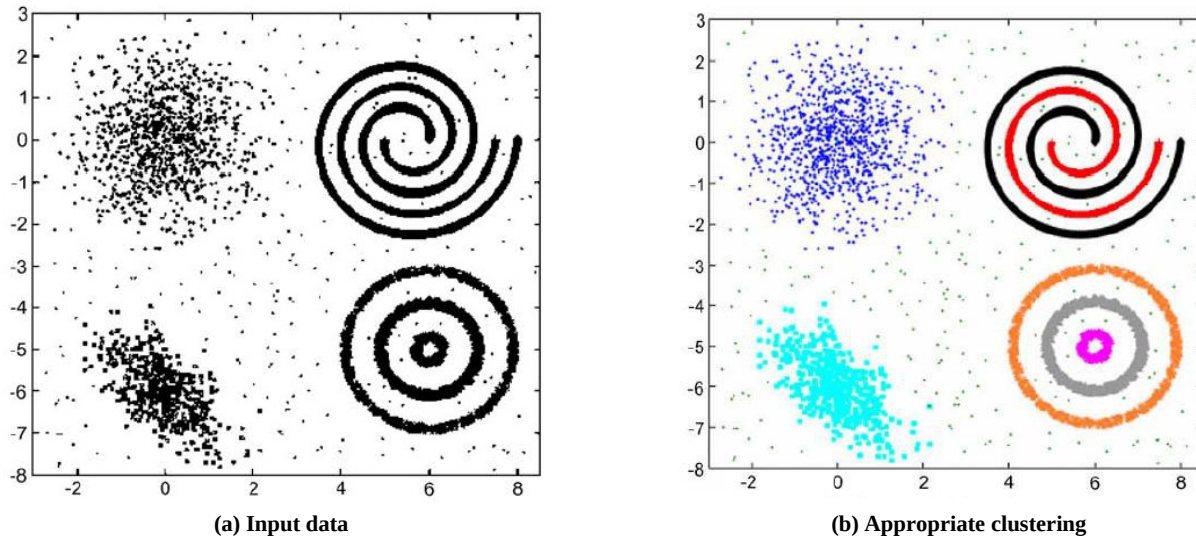


Figure 2-2. Cluster Diversity

Seven clusters of different shapes, sizes, and densities are indicated by seven different colors. Although these clusters are apparent to a data analyst, none of the available clustering algorithms may be able to discover all of them.

Clustering is a common data-mining tool that extracts groups having maximum similarity among elements within the same group and minimum similarity to elements in other groups. Pattern recognition and data analysis with predictive modeling are concerned with learning from training data so that the behavior of previously unseen test data can be predicted. This is generally called learning. A clear distinction is often made between supervised learning, namely classification, and unsupervised learning, namely clustering [5].

Clustering is more difficult and more challenging than classification. It differs from classification because clustering does not attempt to predict or estimate sample values or assign samples to predefined classes. Instead, clustering algorithms seek a partition of all samples into homogeneous subgroups or clusters such that similarity among samples within a cluster is maximized and similarity to samples outside the cluster is minimized. Similarity or dissimilarity is commonly defined through a distance measure. Clustering is therefore an unsupervised classification process in which classes have not been defined in advance.

The purpose of clustering is to provide the final consumer of information with an appropriate view of the events occurring in a database. Another application is to identify data items that differ significantly from the remaining data.

Examples include reducing the dimensionality of a dataset containing hundreds of features, dividing financial behavior into low-risk and risky categories for auditing, and clustering genes when many exhibit similar behavior.

Clustering is sometimes performed as a data-preprocessing stage in the data-mining process. The resulting clusters are then used as inputs to algorithms such as neural networks.

2-3-4-3. Dependency Modeling

Dependency modeling in data mining is the process of finding relationships among features. Its most common business application is known as market-basket analysis or affinity analysis, in which the data-mining process seeks rules that describe relationships among several features.

Examples of association rules or dependency analysis include determining which products are purchased together or not purchased together in a store, identifying combinations in which new medicines may produce dangerous side effects, and examining how Internet users reach desired pages by tracking the links they follow.

2-3-4-4. Estimation

Estimation is the assessment of a cause-and-effect relationship among a set of variables, where one variable is regarded as the consequence of the other variables and their relationships are estimated. Estimation resembles classification, except that target variables have numeric values rather than discrete categories.

Models are constructed from complete samples. For new observations, target-variable values may then be estimated from attributes such as age, sex, weight, and blood sodium level. The relationship between blood pressure and predictor values in a training set helps construct the predictive model, which can then be applied to new samples. Estimating students' postgraduate grade-point averages from their undergraduate averages is another example of estimation.

Statistical analysis provides many estimation methods, including point estimation, confidence-interval estimation, simple linear regression, correlation, and multiple regression.

2-3-4-5. Description

Researchers and analysts sometimes attempt to find methods for describing patterns. Data may be defined by classes or concepts. For example, in an electronics store, a class of items offered for sale may include computers and printers, while the customer concept may include cash-paying and credit-paying customers. This method can describe unique classes and concepts in a concise, compact, and precise form.

2-3-5. Model Interpretation and Result Extraction

In many cases, data-mining models must assist humans in decision-making. These models must therefore be interpretable. It should be noted that model accuracy and the accuracy with which the model can be interpreted are often in tension. Simple models are usually more interpretable, but their accuracy may be lower.

Modern data-mining techniques are expected to provide more accurate results. Interpreting these models, however, becomes another major issue and is treated as a separate operation using specialized techniques for evaluating results.

A user does not want hundreds of pages of numerical results, may not be able to understand them, and cannot readily summarize or translate them into suitable decisions. The iterative data-mining process, including all of the stages described above, is shown in Figure 2-3.

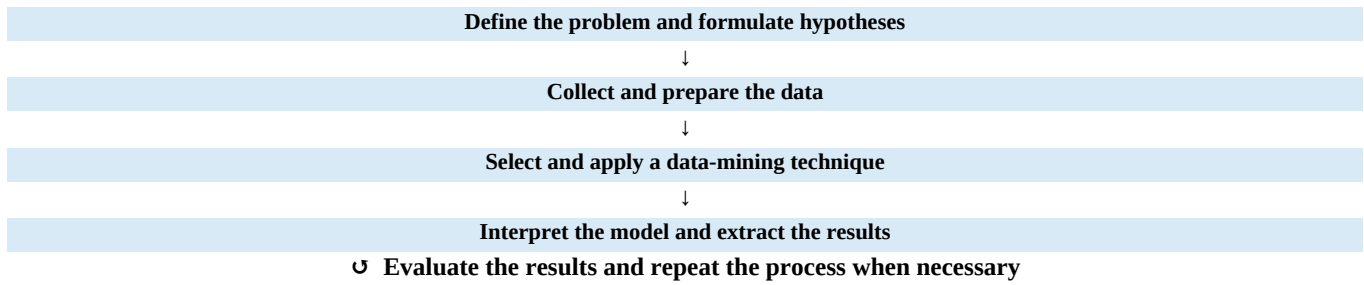


Figure 2-3. Iterative Data-Mining Process

Successful use of data-mining techniques requires an appropriate understanding of the system. The power of a data-mining method alone is not sufficient. Unless the data are collected and preprocessed correctly, the results of a data-mining model will not be valid.

2-4. Large Datasets

With the advent of the digital-information age, the increasing volume of data has become a major issue. Under such conditions, analyzing very large datasets appears difficult. Today, large databases containing growing volumes of digital information are common. Credit-card and banking records, medical-office information, telephone-call patterns, and many other applications continuously generate digital records that are stored in very large commercial databases.

Scientific instruments and equipment can also generate vast volumes of information in a very short time and store them on computers. The maturation of information technology and the development of the Internet have caused both information resources and data-storage centers to expand.

The principal problem is that the size and dimensionality of the data are too large for manual exploration or even for semi-automatic computer-based analysis. A scientist or business manager may work effectively with several hundred or several thousand records, but exploring millions of samples with tens or hundreds of features is a different problem.

Consider, for example, the analysis of terabytes of spatial imagery with a resolution of 32,040 by 32,040 pixels, or a database containing patients' genetic information. In theory, large datasets can provide stronger results, but in practice they create many difficulties.

A survey concerning the vast quantity of information available to businesses and economic organizations indicates that:

- 60% of managers believe that the volume of information in their workplace has increased.
- 80% believe that the situation is becoming worse.
- More than 50% disregard part of this information when making decisions.
- 48% retain the information for future use.
- 60% believe that the cost of collecting the information exceeds the value of retaining it.

In theory, most data-mining techniques should perform better when applied to large datasets. Such datasets have substantial potential for yielding valuable information. If data mining is regarded as a search through a space of possibilities, a large dataset offers many more possibilities to enumerate and evaluate.

In practice, however, the potential of large datasets for extracting valuable information is offset by the complexity of searching them. In addition to the computational complexity of data-mining algorithms, extremely long searches may increase the risk of finding simple solutions that fit the current dataset but fail to satisfy future expectations.

In modern multimedia environments with extensive Internet infrastructures, many different types of data are generated and stored digitally. To prepare suitable data-mining methods, the characteristics and simpler forms of datasets must first be analyzed. The first step is therefore to classify data according to how computers represent and use them. Data used in data mining are generally classified as structured, semi-structured, or unstructured.

Most commercial databases contain structured data composed of defined fields with numerical or alphanumeric values. Scientific databases, however, may contain all three categories of information.

Electronic images of business documents, medical reports, and maintenance notebooks are examples of semi-structured data. Most web-based documents also belong to this category. An example of unstructured data is video recorded by security cameras in large stores. Today, visual information and multimedia records of events or processes are widely used because hardware costs have decreased. Extracting and organizing information from such data requires substantially heavier processing.

Structured data are often called traditional data, whereas semi-structured and unstructured data are known as nontraditional or multimedia data. Most current data-mining methods and available commercial tools have been applied to traditional data. Nevertheless, data-mining tools for multimedia data, including interfaces that convert these data into structured form, are developing rapidly [6].

A standard data-mining model for structured data consists of a set of observations or objects. These objects are evaluated according to criteria called features. Structured data for data-mining problems are commonly displayed as a table. The columns contain the features of the samples, and the rows contain the values of those features for individual samples. A simple graphical representation of a dataset and its characteristics is shown in Figure 2-4.

Modern computers and software tools support the processing of datasets containing millions of samples and hundreds of features. Large datasets containing mixed data types provide an important environment for the application of data-mining techniques.

Dataset

Sample	Feature 1	Feature 2	...	Feature n
Sample 1	x11	x12	...	x1n
Sample 2	x21	x22	...	x2n
Sample 3	x31	x32	...	x3n
Sample 4	x41	x42	...	x4n
Sample 5	x51	x52	...	x5n

Figure 2-4. Tabular Representation of a Dataset

The quality of data is essential to the data-mining process. The quality assessment of a dataset is normally performed during preprocessing. If the data do not accurately represent the system under study, no data-mining technique can produce a valid and useful model. Data quality therefore has a profound effect on the resulting model and the overall system.

A careful examination of the data should identify missing values, invalid records, inconsistent coding, duplicate observations, noise, and outliers. Appropriate cleaning and transformation should then be performed before model construction. The selected representation must also preserve the information required by the intended analysis.

Thus, preparing high-quality data is not merely a preliminary technical task. It is a fundamental requirement for reliable knowledge discovery and for the correct interpretation of subsequent results.

Chapter 3

Research Background

3-1. Introduction

Clustering is a technique for grouping a set of data items into subsets or clusters. The members of each cluster are more similar to one another than to members of other clusters. Similarity is usually determined from the values of the features that describe the objects. The central objective is therefore to obtain high intra-cluster similarity and low inter-cluster similarity.

Cluster analysis has long been used in statistics, pattern recognition, machine learning, information retrieval, image processing, biology, medicine, marketing, and many other scientific and engineering fields. Because the classes are not known in advance, clustering is regarded as an unsupervised learning problem.

A cluster may be described as a collection of objects that are close to one another according to a selected distance or similarity measure. However, there is no single universal definition of a cluster. Compact spherical groups, elongated groups, density-connected regions, and hierarchically nested groups may all be meaningful in different applications. Consequently, the quality of a clustering result depends on the representation of the data, the similarity measure, the algorithm, and the intended application.

Clustering methods may be divided into hierarchical, partitional, density-based, graph-based, model-based, and fuzzy approaches. Different methods may produce different partitions of the same dataset. The selection of an appropriate method therefore requires both knowledge of the data and an understanding of the assumptions of the algorithm.

Figure 3-1 illustrates the basic clustering problem. In the original data space, objects are represented only by their measured features. After clustering, the same objects are assigned to groups that reveal an underlying structure.

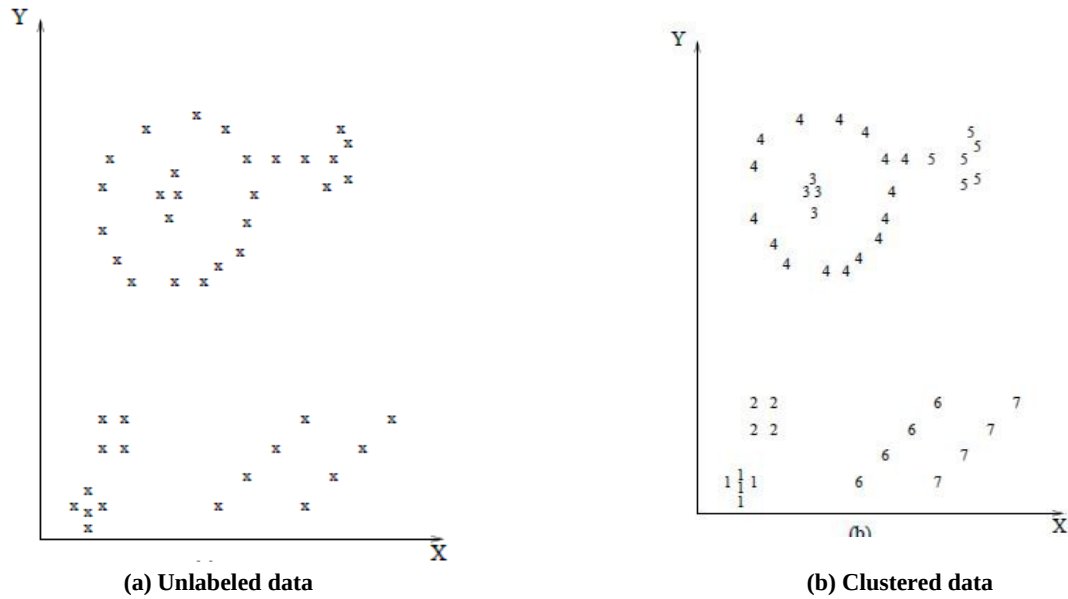


Figure 3-1. Data Clustering

Clustering has two broad purposes. The first is exploratory: discovering structures, generating hypotheses, detecting unusual observations, and identifying prominent features. The second is practical: producing a natural classification, compressing data, simplifying subsequent processing, or preparing data for another learning method.

Clustering may reveal subclasses that were previously unknown. For example, a set of observations nominally assigned to one class may contain several internally distinct groups. Discovering these groups can improve the understanding of the data and may support more accurate classification or prediction in later stages.

3-2. The Need for Clustering

The rapid growth of digital data has made automated organization essential. Large datasets cannot be examined record by record, and their hidden structures are often not evident from summary statistics alone. Clustering provides a means of reducing this complexity by representing many observations through a smaller number of coherent groups.

Clustering is useful when reliable class labels are unavailable, expensive to obtain, or inherently ambiguous. It can expose the natural organization of data before a supervised model is designed. It is also useful for anomaly detection, because observations that do not belong well to any major group can be identified as potential outliers.

In information retrieval, clustering groups similar documents and improves browsing. In image processing, it can divide an image into homogeneous regions. In marketing, it identifies groups of customers with similar behavior. In biology, it groups genes or organisms according to measured characteristics. In workforce management, it can reveal groups with similar skills or performance profiles.

The usefulness of a clustering result is always related to its objective. The same observations can be clustered differently when the analyst emphasizes different features. Thus, there may be several valid partitions of a dataset rather than one uniquely correct solution.

3-3. The Clustering Process

The clustering process generally consists of four closely related stages: feature selection or feature extraction, selection or design of the clustering algorithm, cluster validation, and interpretation of the results. Feedback among these stages is essential because the outcome of one stage may require revision of an earlier decision.

Feature selection chooses the most informative attributes from the available candidates. Feature extraction creates new attributes from the original measurements. Both operations aim to preserve the information needed to distinguish groups while reducing irrelevant variation and computational cost.

The clustering algorithm then organizes the represented objects according to a similarity criterion. The resulting partition must be validated to determine whether the groups are meaningful and stable. Finally, the clusters are interpreted using domain knowledge and the purpose of the analysis.

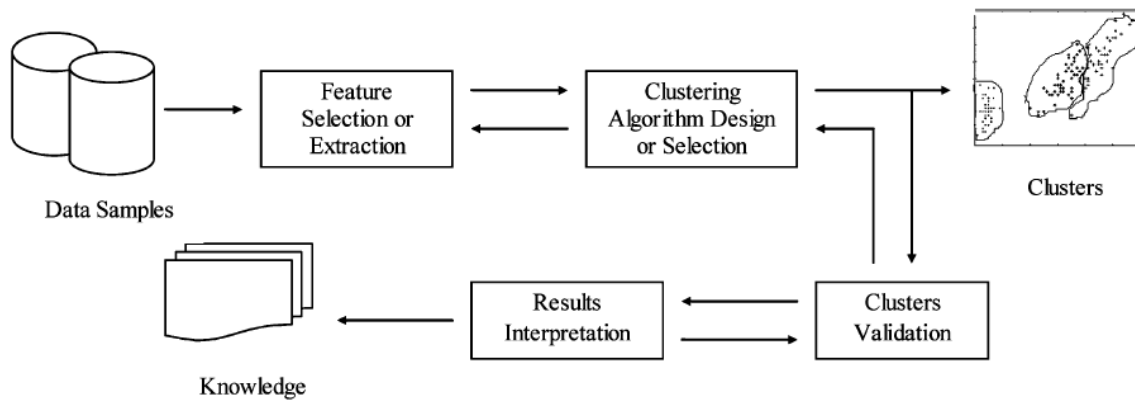


Figure 3-2. Clustering Procedure

As shown in Figure 3-2, raw samples first undergo feature selection or extraction. A suitable representation and similarity measure are then supplied to the clustering algorithm. The resulting clusters are evaluated, interpreted, and, when necessary, fed back to revise the selected features, parameters, or algorithm.

The similarity measure has a direct effect on the final partition. Euclidean distance is commonly used for numerical data, but other measures may be required for binary, categorical, sequential, spatial, or relational data. Scaling and normalization are important because features with larger numerical ranges can dominate the distance calculation.

The choice of algorithm also depends on the expected cluster structure. K-Means is appropriate for compact groups and numerical data, whereas hierarchical, density-based, graph-based, fuzzy, or model-based methods may be more suitable for other structures. No single algorithm performs best for every dataset.



Figure 3-3. Finding Subclasses in Data Using Clustering

Figure 3-3 shows how clustering can divide a nominal class into meaningful subclasses. Such subclass discovery is useful when a single label conceals several distinct patterns.



Figure 3-4. Merging Data in Hierarchical Clustering

Figure 3-4 illustrates agglomerative hierarchical clustering. Each object begins as a separate cluster. At each stage, the two closest clusters are merged. Repeating this operation produces progressively larger groups and ultimately a hierarchy that can be represented by a dendrogram.

The reverse procedure is divisive hierarchical clustering. It begins with all objects in one cluster and repeatedly divides groups until a stopping condition is satisfied. Agglomerative methods are more common because they are generally simpler to implement.

A clustering system includes several design decisions. The analyst must determine the data representation, the objective of grouping, the number of clusters, the similarity measure, the algorithm, and the criterion used to evaluate the result. These decisions are interdependent and cannot be selected reliably without considering the application.

3-4. User Dilemmas

The user of a clustering method faces several fundamental dilemmas. What constitutes a cluster? Which attributes should represent the objects? Should the features be normalized or weighted? How many clusters are present? Which algorithm is suitable? How should validity be measured? Different answers can lead to substantially different results.

3-4-1. Data Representation

Data representation is one of the most important factors in clustering. A good representation emphasizes differences that are relevant to the problem and suppresses irrelevant variation. Poorly selected features may cause genuine groups to overlap or may create artificial separation.

The selected properties may be directly measured features or new features produced by transformation. Domain knowledge is often necessary to decide which representation is meaningful. Dimensionality reduction can improve efficiency and reduce noise, but important information must not be discarded.

For relational data, objects may be represented by a proximity matrix rather than by vectors. Kernel and radial-basis representations can also transform nonlinear structures into a space in which groups are easier to separate.

3-4-2. Purpose of Grouping

The purpose of grouping determines which similarities are important. In a dataset describing animals, for example, a partition based on physical appearance may differ from one based on activity or habitat. Both may be correct for their respective objectives. Feature weighting is therefore not merely a numerical adjustment; it expresses the intended meaning of the clustering task.

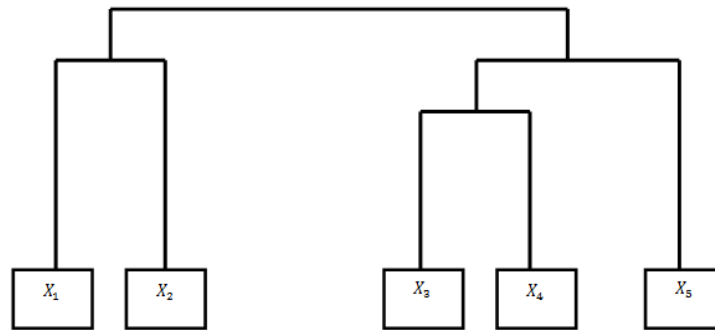


Figure 3-5. A Typical Sequence of the First Three Agglomerative Steps

The first merges in an agglomerative procedure are determined by the smallest pairwise or inter-cluster distances. Different linkage criteria, such as single linkage, complete linkage, or average linkage, can therefore generate different hierarchies.

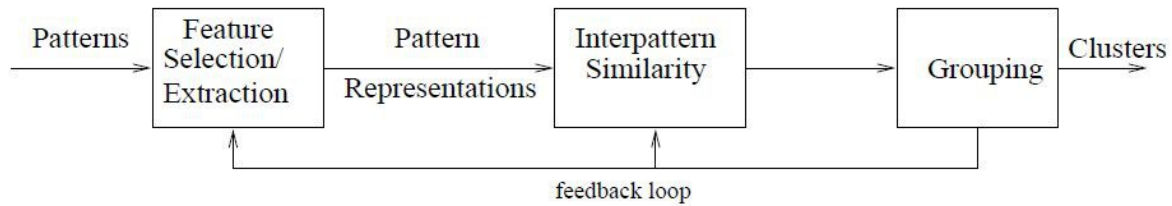


Figure 3-6. A Classification of Clustering Techniques

Clustering methods may be characterized along several dimensions. They may be hierarchical or partitional, agglomerative or divisive, hard or fuzzy, deterministic or stochastic, incremental or non-incremental, and monothetic or polythetic. In hard clustering each object belongs to one cluster, whereas fuzzy clustering assigns degrees of membership. Monothetic methods use one feature at a time, whereas polythetic methods use several features simultaneously.

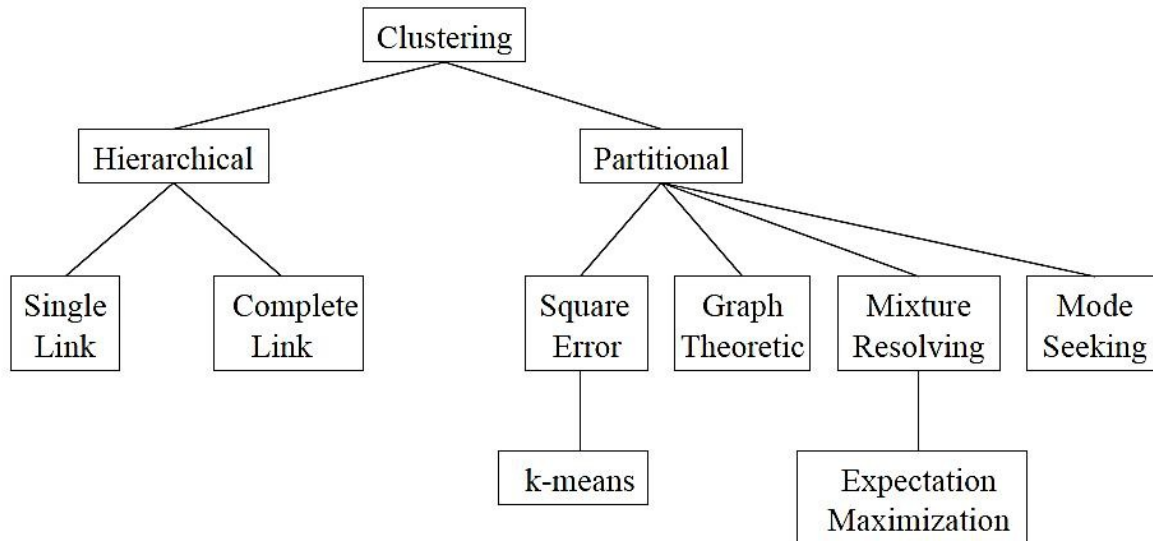


Figure 3-7. Single-Linkage Clustering

Single-linkage clustering defines the distance between two clusters as the minimum distance between any pair of objects, one from each cluster. This method can identify elongated and irregular groups, but it is sensitive to chaining, where intermediate observations connect otherwise separate clusters.

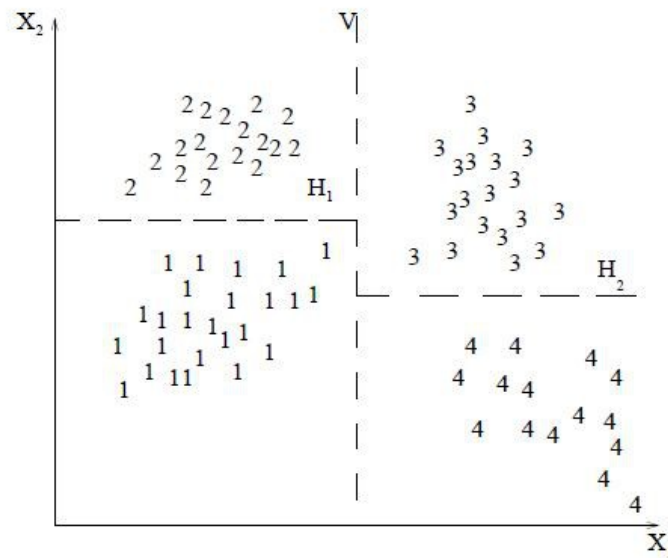


Figure 3-8. Importance of an Appropriate Data Representation

The figure demonstrates that the success of clustering depends strongly on the representation. In the original feature space, groups may overlap and be difficult to separate. After an appropriate transformation or similarity representation, the same observations can form compact and clearly separated clusters.

Because clustering is unsupervised, the discovered labels are not automatically reliable. Cluster validity must therefore be examined using internal criteria, comparisons among alternative partitions, external information when available, and stability under changes in samples or parameters.

The representation selected for the data can make a cluster structure either visible or obscure. A clustering algorithm cannot recover information that has been removed by an unsuitable representation. Accordingly, feature selection, feature weighting, normalization, and transformation must be treated as integral parts of the clustering problem.

When the data contain features with different units or ranges, direct use of Euclidean distance may give excessive influence to large-scale variables. Standardization or domain-based weighting is therefore required. However, indiscriminate normalization can also remove meaningful differences. The correct treatment depends on the application.

A representation that is appropriate for one clustering objective may be inappropriate for another. For example, animals may be grouped according to physical appearance, activity, habitat, or biological relationship. Each objective assigns different importance to the same features and can produce a different but meaningful partition.

Relational representations are useful when objects cannot be described naturally by fixed-length vectors. In such cases, the algorithm works with a similarity or dissimilarity matrix. Kernel functions and radial-basis transformations can also reveal nonlinear group structures that are not separable in the original feature space.

Figure 3-9 demonstrates that changing feature weights changes the resulting partition. This sensitivity means that clustering results should always be interpreted together with the representation and similarity measure from which they were obtained.

3-4-2. Purpose of Grouping

The purpose of grouping should be specified before selecting the algorithm. If the objective is data compression, compact representative clusters may be preferred. If the objective is anomaly detection, the method must preserve small or isolated groups. If the objective is taxonomy, a hierarchy may be more informative than a single flat partition.

The desired result may be a partition, a hierarchy, a probabilistic model, or a set of overlapping groups. Hard clustering assigns every object to exactly one cluster. Fuzzy or probabilistic clustering permits partial membership, which may better describe observations located near cluster boundaries.

The usefulness of a partition is not determined solely by numerical compactness. A mathematically compact solution may have little domain meaning, whereas a slightly less compact partition may correspond to important real categories. Domain knowledge is therefore necessary when evaluating clustering results.

3-4-3. Number of Clusters

Determining the number of clusters is one of the most difficult problems in cluster analysis. Many algorithms require the number of clusters to be specified in advance, although the correct value is usually unknown. Running an algorithm for several values of K and selecting the best result according to a validity criterion is a common approach.

Model-selection criteria such as Minimum Message Length (MML) and Minimum Description Length (MDL) balance model fit against complexity. A model with too few clusters cannot represent the data adequately, while one with too many clusters describes noise and loses generality.

In mixture-model clustering, the number of components can be selected by comparing the likelihood and complexity of competing models. Gaussian Mixture Models are a common example. Each component represents a probability distribution, and the combined model describes the overall density of the data.

An automatic procedure may begin with many candidate components and then merge or remove components when doing so reduces the MML or MDL criterion. Alternatively, models with different values of K may be estimated independently and compared. The selected number should remain stable under reasonable changes in initialization and sampling.

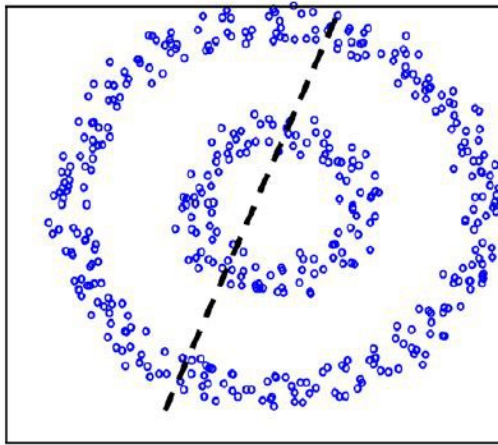
The cluster count identified by a numerical criterion is not necessarily the only meaningful count. Hierarchical data can support several valid levels of granularity. A biological dataset, for instance, may be divided into broad families at one level and into species at another. The required resolution depends on the intended use.

Noise and outliers complicate cluster-count estimation. Small groups may represent genuine rare classes or may consist only of random observations. Density-based methods can explicitly label low-density observations as noise, whereas partitional methods generally force every observation into a cluster.

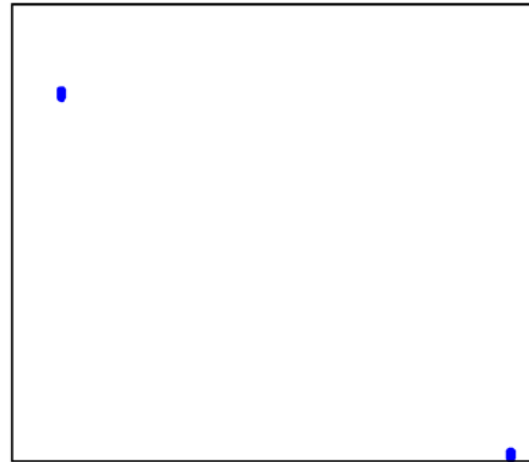
Figure 3-10 illustrates automatic cluster-number selection for synthetic data generated from Gaussian mixtures. Candidate solutions with different component counts are compared to determine the model that best balances fit and complexity.

3-4-4. Cluster Validity

Cluster validity evaluates whether a discovered partition reflects meaningful structure rather than an artifact of the algorithm or initialization. Validity criteria are commonly divided into internal, relative, and external measures.



(a) Original data distribution



(b) Alternative representation

Figure 3-9. Effect of Feature Representation and Weighting on Clustering Results

Internal criteria use only the data and the obtained partition. They generally reward compact clusters that are well separated from one another. Examples include the sum of squared errors, silhouette-based measures, and indices based on within-cluster and between-cluster distances.

Relative criteria compare several partitions produced by different parameter values or algorithms and select the best one. External criteria compare the obtained clusters with known class labels or another accepted reference partition. External validation is possible only when trustworthy reference information is available.

Stability is another important aspect of validity. A meaningful clustering should not change radically when the data are slightly perturbed, when a subset is resampled, or when the algorithm is initialized differently. Large variation across repeated runs indicates that the structure is weak or the method is unstable.



Figure 3-10(a). Gaussian-Mixture Dataset and Candidate Clustering

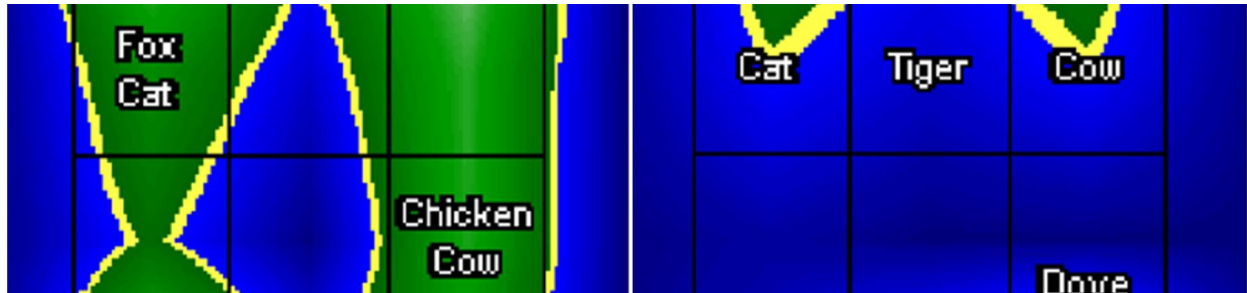


Figure 3-10(b). Alternative Numbers of Gaussian Components



Figure 3-10(c). Model Selection for the Number of Clusters

Figure 3-10 compares synthetic Gaussian-mixture data under alternative component counts. A satisfactory selection criterion should identify the underlying structure without splitting a true component unnecessarily or combining clearly distinct components.

Cluster labels themselves are arbitrary. Two algorithms may produce equivalent partitions while assigning different numerical labels to the groups. Comparisons must therefore account for label permutation rather than directly comparing label numbers.

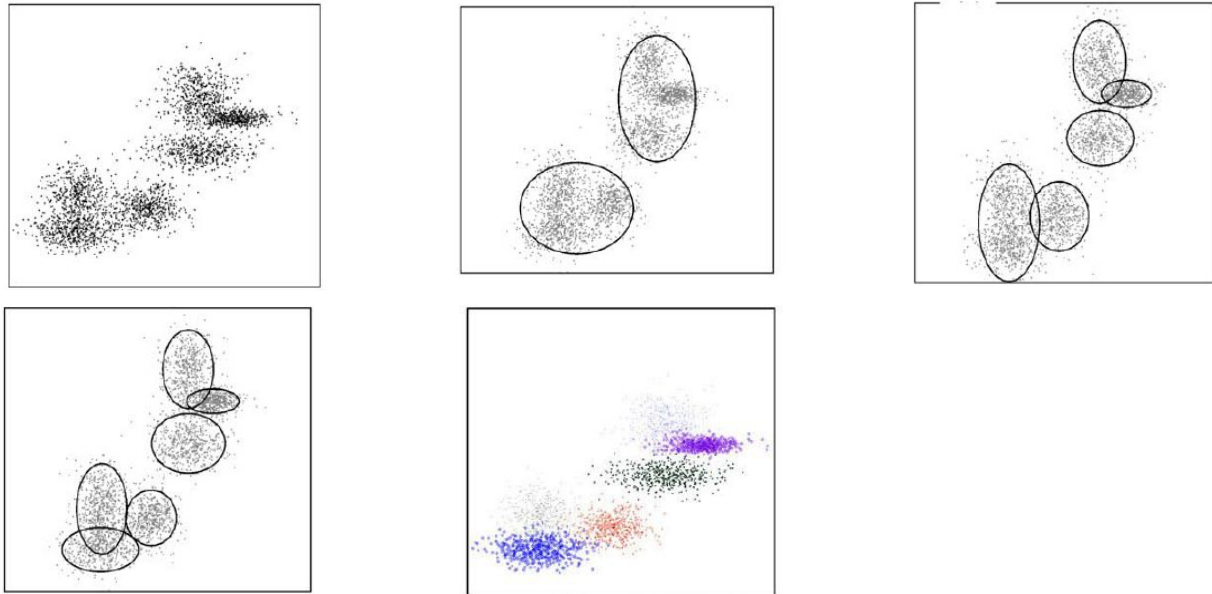


Figure 3-10(d). Candidate Gaussian-Mixture Models with Different Numbers of Components

The candidate models illustrate underfitting, suitable fitting, and overfitting. Too few components merge distinct distributions, whereas too many components divide a coherent distribution into unnecessary subgroups. The preferred model is the simplest model that adequately explains the observed structure.

Validity should be assessed with more than one criterion whenever possible. A single index may favor a particular cluster shape or algorithm. Visual inspection, domain interpretation, external evidence, and stability analysis provide complementary information.

3-5. Clustering Techniques

Clustering algorithms can be organized into several broad families. Hierarchical methods construct nested groups; partitional methods directly divide the data into a fixed number of clusters; density-based methods identify connected high-density regions; graph-based methods use edges among neighboring objects; and model-based methods estimate a statistical model for each cluster.

3-5-1. Hierarchical Clustering Algorithms

Hierarchical clustering produces a sequence of nested partitions that can be represented as a tree or dendrogram. Agglomerative methods begin with each observation in a separate cluster and successively merge the most similar clusters. Divisive methods begin with all observations together and successively split clusters.

The distance between clusters is defined by a linkage rule. Single linkage uses the minimum pairwise distance, complete linkage uses the maximum pairwise distance, and average linkage uses the average distance between pairs. Minimum-variance methods merge the pair that causes the smallest increase in within-cluster variation.

Hierarchical methods are flexible because a dendrogram can be cut at different levels to obtain different numbers of clusters. They can also use many kinds of proximity measures. However, once an agglomerative merge or divisive split has been made, it is generally not reversed. Early errors can therefore propagate through the hierarchy.

Another difficulty is choosing a stopping level. Large changes in linkage distance may indicate a natural cut, but such changes are not always present. The final level should be selected using validity criteria and domain knowledge.

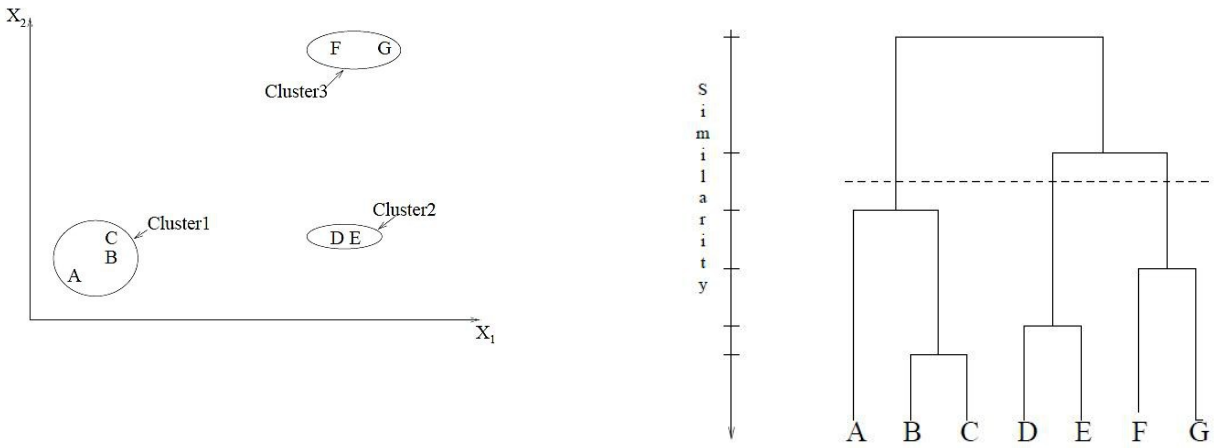


Figure 3-11. Hierarchical Clustering of Points into Three Clusters

The left diagram shows a spatial arrangement of observations, while the dendrogram on the right records the sequence and distance of merges. Cutting the dendrogram at a selected height produces three clusters. Lower cuts create more detailed groups, and higher cuts create broader groups.

Single-linkage methods are capable of detecting non-spherical clusters but may suffer from chaining. Complete linkage generally produces compact clusters but is sensitive to outliers. Average linkage provides a compromise between these behaviors. Ward-type minimum-variance methods favor compact groups and are often effective for standardized numerical data.

The computational requirements of hierarchical clustering depend on implementation and dataset size. A complete proximity matrix requires substantial memory for large datasets. Efficient nearest-neighbor and sparse-graph implementations can reduce this cost, but hierarchical methods are still commonly applied to small or medium-sized datasets or to summaries obtained through preliminary clustering.

A dendrogram provides more information than a single partition because it displays relationships among clusters at several levels. This property is particularly useful for taxonomy, document organization, biological analysis, and exploratory studies in which the appropriate level of detail is not known beforehand.

Nevertheless, the visual structure of a dendrogram must be interpreted cautiously. The ordering of leaves can vary without changing the hierarchy, and apparent horizontal proximity between leaves does not necessarily indicate a small original distance. The meaningful information is the branching structure and merge height.

The output of hierarchical clustering may also be used to initialize a partitional algorithm. The hierarchy supplies an informed initial partition, after which an optimization procedure such as K-Means refines the cluster centers and memberships. Such hybrid strategies combine the exploratory advantages of a hierarchy with the local optimization ability of partitional methods.

3-5-2. Partitional Clustering

Partitional methods divide the dataset directly into K non-overlapping clusters. They usually optimize an objective function that measures cluster compactness or model fit. Because the optimization problem is generally non-convex, different initializations may produce different local optima. Multiple runs and careful initialization are therefore required.

K-Means is the most widely used partitional method. It alternates between assigning each observation to the nearest center and recalculating each center as the mean of its assigned observations. The process continues until memberships or centers no longer change significantly. K-Means is simple and efficient but is sensitive to initialization, scaling, outliers, and non-spherical cluster shapes.

Numerous studies provide guidance for designing partitional clustering methods. These methods generally produce the final clusters by optimizing an objective function either locally over subsets of patterns or globally over the entire dataset. Exhaustively examining all possible partitions to obtain the optimum objective-function value is computationally infeasible. In practice, algorithms are therefore run from several initializations, and the best solution obtained is selected as the final clustering.

3-5-2-1. Squared-Error Algorithm

One of the most widely used objective functions in partitional clustering is the squared-error criterion, which is appropriate for compact, well-separated clusters. For a dataset X partitioned into k clusters, the criterion is defined as:

$$L^2(X, C) = \sum_{j=1}^k \sum_{i=1}^{n_j} \|x_i^{(j)} - c_j\|^2$$

where $x_i^{(j)}$ is the i -th pattern belonging to the j -th cluster, n_j is the number of patterns in that cluster, and c_j is the centroid of the j -th cluster. Minimizing this expression reduces within-cluster dispersion.

K-Means is a simple and common clustering algorithm that uses the squared-error criterion [39]. The method begins with an initial specification of the number of clusters and repeatedly reassigns samples and updates cluster centers until no sample changes cluster or the objective function no longer changes appreciably. Because of its simple implementation and approximately linear dependence on the number of samples, K-Means is widely used. Its principal weakness is sensitivity to initial cluster centers, which can cause convergence to a local minimum.

Footnote 16: Local Minimum

The general squared-error clustering procedure is:

1. Select k clusters randomly.
2. Assign each sample to the nearest cluster centroid and recompute the centroids.
3. Repeat Step 2 until cluster memberships remain unchanged.
4. Split or merge clusters using suitable heuristic information, when required.

Many variants of K-Means have been reported [29]. Some improve initialization in an attempt to approach the global minimum. Others permit the resulting clusters to be divided or merged. Two clusters are commonly merged when the distance between their centroids is below a threshold and divided when their dispersion exceeds another threshold.

A standard K-Means procedure can be stated as follows:

1. Choose k initial cluster centers, usually by selecting k samples.
2. For each sample, identify the nearest center. After this step, all samples are assigned to one of the k clusters.
3. For each cluster, compute and update the centroid.
4. Repeat the assignment and update steps until the termination condition is satisfied.

3-5-2-2. Graph-Theoretic Clustering

A well-known graph-theoretic divisive method constructs a Minimum Spanning Tree (MST) over the input data and removes edges having the greatest lengths, thereby partitioning the data into new clusters [30]. Hierarchical clustering is closely related to graph-theoretic clustering. Single-linkage clustering corresponds to subgraphs of the MST, while complete-linkage clustering is related to complete subgraphs [31]. Graph-oriented methods for non-hierarchical structures and overlapping clusters have also been presented [32].

3-5-3. Nearest-Neighbor Clustering

Because proximity plays a central role in clustering algorithms, the nearest-neighbor method can be regarded as one of the most elementary clustering processes. One form of this process is described in [33].

For each sample, the distance to its nearest neighboring sample is computed. If this distance is less than a specified threshold, the two samples are assigned to the same cluster. The procedure continues until all samples have been assigned labels. The threshold determines the granularity of the result: a small threshold produces many small clusters, whereas a large threshold can merge distinct structures.

Nearest-neighbor clustering can preserve irregular shapes because membership is determined from local relationships rather than from distance to a global centroid. However, the method is sensitive to noise, variations in density, and the selected threshold. Efficient nearest-neighbor search is also important for large or high-dimensional datasets.

3-6. Cluster Ensembles

A cluster ensemble combines several different partitions of the same dataset to obtain a single consensus partition. Individual partitions may be produced by different algorithms, parameter values, initializations, samples, or feature subsets. The objective is to obtain a more reliable and stable result than that produced by a single clustering run.

A cluster-ensemble system has two main components: generation of diverse base partitions and application of a consensus function. Diversity is necessary because combining nearly identical partitions provides little additional information. However, the base partitions must still contain useful structure; completely random partitions cannot produce a meaningful consensus.

3-6-1. Diversity

Diversity among base clusterings can be generated in several ways. A stochastic algorithm can be run repeatedly with different random initializations. Deterministic algorithms can be applied after changing their parameters, distance measures, or requested cluster counts. Resampling can generate different subsets of observations, while random subspace methods use different subsets or transformations of features.

Heterogeneous ensembles apply several clustering algorithms, such as K-Means, hierarchical methods, graph methods, or mixture models. Homogeneous ensembles use one base algorithm but vary initialization, parameters, samples, or features. Both strategies seek complementary errors so that the consensus process can reinforce stable structure and suppress unstable assignments.

The amount of diversity must be controlled. If every partition is nearly the same, the ensemble offers little improvement. If diversity is excessive, agreement disappears. Successful ensemble construction therefore balances quality and diversity.

3-6-2. Consensus Function

Let $\Pi = \{\pi_1, \pi_2, \dots, \pi_n\}$ denote a set of generated partitions. A consensus function maps this collection to a final partition π^* . The function must account for the fact that cluster labels are arbitrary and cannot be compared directly across partitions.

One common method constructs a co-association matrix. Each matrix entry records how often two samples occur in the same cluster across the ensemble. A conventional clustering algorithm is then applied to this matrix. Other approaches maximize mutual information, partition a hypergraph, fit a finite mixture model, or relabel clusters before voting.

The quality of the final consensus depends on the diversity and quality of the base partitions, the number of partitions, and the assumptions of the consensus function. Consensus clustering often improves stability and reduces sensitivity to initialization, although it introduces additional computational cost.

3-7. Summary and Conclusions

This chapter reviewed the principal concepts and methods of cluster analysis. Data representation, similarity measurement, the number of clusters, cluster validity, and algorithm selection were identified as central issues. Hierarchical, partitional, graph-based, nearest-neighbor, and ensemble methods each have advantages and limitations. No single technique provides the best accuracy for every application, and exact determination of the number and structure of clusters remains difficult.

Chapter 4

Proposed Data-Clustering Method

4-1. Introduction

Nature-inspired computation has provided effective methods for solving complex optimization problems. These methods imitate collective behavior, evolution, adaptation, and learning in biological or social systems. Their population-based search mechanisms can explore large solution spaces and reduce dependence on a single initial solution.

Metaheuristic algorithms do not guarantee the exact global optimum, but they can obtain acceptable solutions to problems for which exhaustive search is impractical. Genetic algorithms, particle swarm optimization, ant-colony optimization, artificial bee-colony methods, and bacterial foraging are examples of this class.

This thesis uses the Bacterial Foraging Optimization Algorithm together with the Cultural Algorithm. Bacterial foraging supplies the population-based search behavior, while the Cultural Algorithm stores and communicates knowledge obtained during the search. Their combination is used to improve data clustering.

4-2. Bacterial Foraging Optimization Algorithm

The Bacterial Foraging Optimization Algorithm (BFOA) is inspired by the foraging behavior of bacteria such as *Escherichia coli*. Bacteria move through their environment in response to chemical gradients. Attractive nutrients guide movement toward favorable regions, whereas harmful substances cause movement away from unfavorable regions.

The basic biological processes represented in BFOA are chemotaxis, swarming, reproduction, and elimination-dispersal. During chemotaxis a bacterium alternates between tumbling, which selects a new direction, and swimming, which continues movement in a favorable direction. Through swarming, bacteria communicate and form groups around promising nutrient sources.

After a chemotactic period, healthier bacteria reproduce and less successful bacteria are removed. Elimination-dispersal events randomly relocate some bacteria, helping the population escape local optima and explore new regions of the search space.

4-2-1. Biological Basis and Bacterial Movement

The movement of a bacterium is governed by the chemical concentration of the environment. A positive nutrient gradient encourages continued movement, while an adverse gradient causes a tumble and a change in direction. This behavior produces a balance between local exploitation and global exploration.

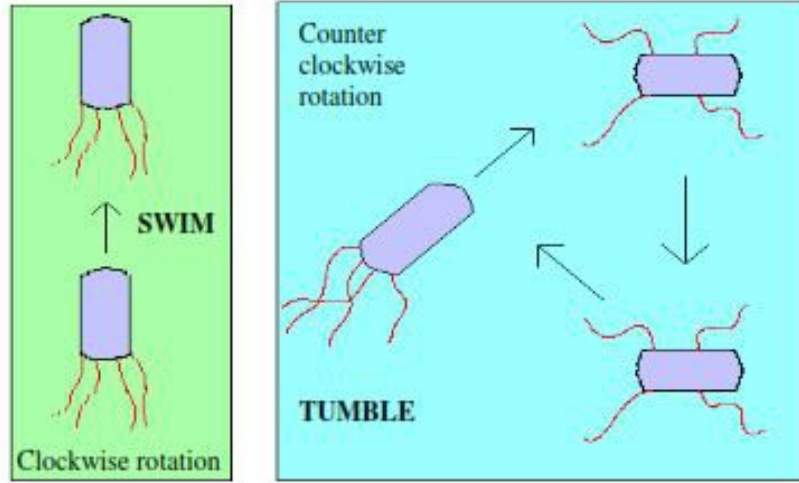


Figure 4-1. Bacterial Movement During Chemotaxis

The interaction of bacteria in a swarm can be modeled by an attraction-repulsion function. Attraction encourages bacteria to gather around good solutions, while repulsion prevents excessive concentration at a single point. The total interaction depends on the positions of all bacteria, the dimensions of the search space, and the attraction and repulsion coefficients.

$$J_{cc}(\theta, P) = \sum_{i=1}^S [-d_a \exp(-w_a \sum_{m=1}^P (\theta_m - \theta_m^i)^2) + h_r \exp(-w_r \sum_{m=1}^P (\theta_m - \theta_m^i)^2)]$$

Here S is the number of bacteria, P is the dimension of the search space, θ is the candidate position, θ^i is the position of bacterium i , and the remaining coefficients control the depth and width of the attractive and repulsive effects. The interaction term is added to the objective function when the health of a bacterium is evaluated.

4-2-2. Operation of the Bacterial Foraging Method

BFOA represents each candidate solution as the position of a bacterium in the search space. The objective-function value determines the amount of nutrient available at that position and therefore the health of the bacterium. The algorithm repeatedly performs chemotaxis, reproduction, and elimination-dispersal.

Chemotaxis. A bacterium tumbles by selecting a random direction and then moves by a step of specified size. If the new position improves the objective value, the bacterium continues swimming in the same direction for a limited number of steps. Otherwise, it tumbles again.

Swarming. The attraction-repulsion interaction among bacteria encourages the population to form groups around promising regions while preserving enough separation to avoid complete collapse.

Reproduction. After a prescribed number of chemotactic steps, the health of each bacterium is computed from its accumulated objective values. The least healthy half of the population is removed, and each bacterium in the healthier half is duplicated at the same position. Population size therefore remains constant.

Elimination-dispersal. After several reproduction cycles, each bacterium is eliminated and randomly dispersed with probability P_{ed} . This operation models sudden environmental changes and helps the search escape local optima.

The principal parameters include the number of bacteria S , search-space dimension P , number of chemotactic steps N_c , swimming length N_s , number of reproduction steps N_{re} , number of elimination-dispersal events N_{ed} , step size $C(i)$, and elimination-dispersal probability P_{ed} .

The process ends when the maximum number of iterations is reached or another termination criterion is satisfied. The best bacterium encountered during the search is retained as the final solution.

BFOA Pseudocode

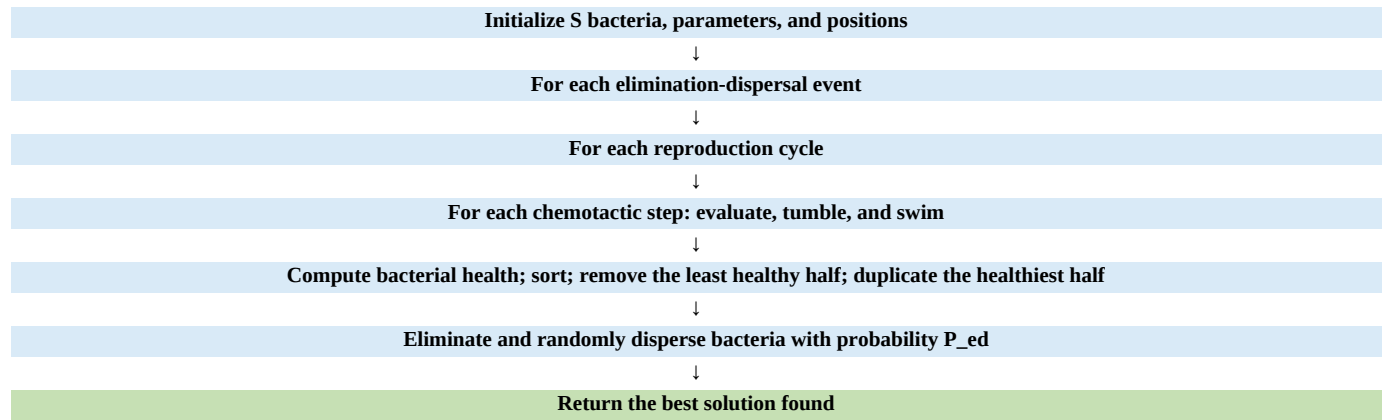


Figure 4-2. Pseudocode of the Bacterial Foraging Optimization Algorithm

4-3. Cultural Algorithm

The Cultural Algorithm models cultural evolution in a society. Unlike a conventional evolutionary algorithm, which stores knowledge only implicitly in the population, the Cultural Algorithm maintains an explicit knowledge repository called the belief space. This knowledge influences the evolution of individuals in the population space.

BELIEF SPACE

Normative Knowledge	Situational Knowledge	Domain Knowledge
Temporal Knowledge	Spatial Knowledge	Search History

↑ **Acceptance Function: selected individuals update beliefs**

↓ **Influence Function: beliefs guide variation and selection**

POPULATION SPACE: individuals, evaluation, selection, reproduction, and mutation

Figure 4-3. Operation of the Cultural Algorithm

The Cultural Algorithm has three principal components: a population space, a belief space, and communication protocols connecting the two. The population evolves through ordinary evolutionary operators. Selected high-quality individuals update the belief space through the acceptance function, while the belief space guides the population through the influence function.

4-3-1. Belief Space

The belief space is a repository of knowledge extracted from the experiences of the population. It preserves information that may be lost when individual members are replaced and can direct the search away from undesirable regions and toward promising regions.

At every generation, an acceptance function selects individuals whose information is permitted to modify the belief space. The influence function then uses the updated beliefs to alter variation, mutation, or the selection of new candidate solutions. Culture and population therefore affect one another continuously.

The belief space can reduce unnecessary exploration by narrowing the plausible search region. However, if it becomes too restrictive, diversity may decrease and the search may converge prematurely. The acceptance and influence mechanisms must therefore balance exploitation of accumulated knowledge with exploration of new regions.

4-3-2. Knowledge Components

Normative knowledge. Specifies acceptable ranges and desirable behavior for the variables of the problem.

Domain knowledge. Contains information and relationships specific to the problem domain.

Situational knowledge. Stores important examples, exemplars, or experiences associated with particular events.

Temporal knowledge. Records the historical sequence of changes and the direction of progress during the search.

Spatial knowledge. Describes the topology of the search space, including promising and undesirable regions.

4-3-3. Communication Protocols

Communication protocols transfer information between the population space and the belief space. They determine which individuals influence culture and how cultural knowledge modifies the population. These protocols control the feedback loop that distinguishes a Cultural Algorithm from an ordinary evolutionary algorithm.

The effect of the population on culture may be static or dynamic. In the static method, a fixed percentage of the highest-fitness individuals is accepted into the belief-space update. In the dynamic method, the number or proportion of accepted individuals changes with the generation and the progress of the search.

Let popsize denote the population size, P_accept the acceptance proportion, and G the generation counter. The acceptance mechanism is selected so that early generations preserve diversity, while later generations give greater influence to high-quality individuals. The exact relation used in the algorithm is given in Equation 4-2.

$$N_accept(G) = f(popsize, P_accept, G)$$

The influence of culture on the population is implemented through an influence function. This function alters individuals so that their candidate solutions move closer to globally or situationally desirable beliefs. Mutation is used to introduce the effect of the belief space while retaining sufficient variation.

A strong influence function accelerates convergence but can reduce exploration. A weak influence function preserves diversity but may not exploit accumulated knowledge effectively. Parameter selection is therefore important for the performance of the Cultural Algorithm.

4-4. Data Clustering by Combining the Cultural Algorithm and BFOA

In the proposed method, every bacterium represents a candidate clustering solution. The genes of a bacterium encode cluster assignments or cluster-related information for the input records. The objective function evaluates the compactness and separation of the resulting clusters.

The Cultural Algorithm stores knowledge obtained from successful bacteria. This knowledge guides the generation and mutation of new candidates, while BFOA supplies chemotactic movement, reproduction, and elimination-dispersal. The hybrid method is therefore intended to combine adaptive memory with population-based exploration.

The general stages of the proposed method are:

1. Preprocess and normalize the input dataset.
2. Generate the initial population of bacteria and encode a candidate clustering in each bacterium.
3. Evaluate every bacterium using the clustering objective function.
4. Initialize the belief space from the best individuals.
5. Repeat until the termination condition is satisfied: update the population through bacterial operations, update the belief space from accepted individuals, and apply cultural influence to mutation and repair.
6. Select the best bacterium as the final clustering.

The Davies-Bouldin index is used as one criterion for evaluating candidate partitions. Lower values indicate clusters that are internally compact and well separated from one another.

Several structural mutation operators are used to improve candidate clusterings. Gene replacement changes the assignment encoded at one position. Gene division separates part of a cluster to create an additional group. Gene merging combines two clusters when the resulting solution is more suitable. Each operator is applied according to its prescribed probability.

After mutation, invalid candidate solutions are repaired so that every record has a valid cluster assignment and the number of clusters remains within the permitted range. Gene transfer allows information from a successful bacterium or from the belief space to be introduced into another candidate.

The fitness of the new population is evaluated, and superior bacteria are selected for the next generation. The belief space is then revised using the accepted individuals. Repeating this interaction allows historical experience to affect future search behavior rather than being discarded when the population changes.

The proposed algorithm preserves diversity through elimination-dispersal and cultural mutation, while reproduction and belief-space influence intensify the search around good solutions. This balance is expected to reduce premature convergence and improve clustering accuracy.

The parameter values used in the experiments include population size, maximum number of iterations, number of reproduction steps, probabilities of replacement, splitting and merging, maximum number of clusters, and the belief-space acceptance rate. Their experimental values are reported in Chapter 5.

4-5. Summary and Conclusions

This chapter described BFOA, the Cultural Algorithm, and their combination for data clustering. BFOA provides chemotaxis, swarming, reproduction, and elimination-dispersal. The Cultural Algorithm adds explicit knowledge storage and communication between the population and belief spaces. In the proposed hybrid algorithm, cultural knowledge guides bacterial search and mutation to obtain compact and well-separated clusters.

The algorithm begins by preparing the data and forming a population of candidate clusterings. Each bacterium is evaluated, and the best bacteria update the belief space. Cultural influence modifies subsequent candidates, while bacterial reproduction preserves successful solutions and elimination-dispersal supports exploration.

The use of several mutation operations enables the encoded clustering structure to change. Replacement changes individual assignments, splitting increases the number of clusters, and merging decreases it. Repair operations maintain valid solutions after these changes.

The proposed approach is designed to address weaknesses of conventional methods such as sensitivity to initialization and entrapment in local optima. The next chapter evaluates the method on standard datasets and compares the results with other clustering algorithms.

The experimental implementation applies the complete hybrid process repeatedly because both BFOA and the Cultural Algorithm are stochastic. Repeated runs permit the best, worst, mean, and standard-deviation values to be reported and provide a more reliable assessment than a single execution.

The comparison focuses on clustering compactness, misclassification error, and stability. The same datasets and evaluation criteria are used for all compared methods so that performance differences can be interpreted consistently.

The following chapter presents the parameter settings, dataset characteristics, experimental procedure, numerical results, and discussion of the proposed Cultural Bacterial Foraging clustering method.

Chapter 5

Experiments and Evaluation of Results

5-1. Introduction

This chapter evaluates the proposed hybrid Cultural Bacterial Foraging clustering algorithm. The experiments were implemented in MATLAB and executed repeatedly because the compared optimization methods are stochastic. Performance is assessed using standard benchmark datasets and compared with K-Means, Genetic Algorithm clustering, Particle Swarm Optimization clustering, and the basic Bacterial Foraging Optimization Algorithm.

5-2. Evaluation Criteria

Clustering quality is evaluated from both internal compactness and external classification agreement. Lower values are preferred for the criteria used in this chapter.

5-2-1. Intra-Cluster Distance

Intra-cluster distance measures the compactness of the obtained clusters. It is calculated as the sum of Euclidean distances between each sample and the centroid of the cluster to which it is assigned:

$$D = \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - c_j\|$$

Here k is the number of clusters, C_j is cluster j , x_i is a sample assigned to that cluster, and c_j is the corresponding centroid. A smaller value indicates more compact clusters.

5-2-2. Error Rate

When the true class labels are available, error rate measures disagreement between the clustering result and the reference classes. Because cluster labels are arbitrary, the obtained clusters are first matched to the reference classes to maximize correct assignments.

$$\text{Error Rate} = (\text{Number of incorrectly clustered samples} / \text{Total number of samples}) \times 100$$

5-3. Experiments

The proposed algorithm and the comparison methods were applied to the Iris, Wine, and Wisconsin Breast Cancer datasets. Each method was run repeatedly under the same evaluation conditions. The reported results include the best, worst, mean, and standard deviation where available.

Table 5-1. Parameter Values of the Proposed Algorithm

Parameter	Symbol	Value
Population size	S	10
Maximum iterations	MaxIter	100
Reproduction interval	Nre	30
Replacement probability	Pr	0.60
Split probability	Ps	0.10
Merge probability	Pm	0.10
Maximum number of clusters	Kmax	10
Belief-space acceptance ratio	Pac	0.20

The benchmark datasets differ in sample size, dimensionality, and number of classes. Their principal characteristics are given below.

Table 5-2. Summary of Dataset Characteristics

Dataset	Samples	Classes	Features
Iris	150	3	4
Wine	178	3	13
Wisconsin Breast Cancer	683	2	9

Iris contains measurements of sepal and petal dimensions for three iris classes. Wine contains thirteen chemical attributes for three cultivars. The Wisconsin dataset contains nine diagnostic attributes and two classes. Before clustering, feature values were normalized to prevent attributes with large numerical ranges from dominating Euclidean distance.

For each dataset, the proposed method was compared with the reference algorithms using intra-cluster distance and error rate. The best solution from repeated runs was retained, while the standard deviation was used to assess stability.

Table 5-3. Comparison Based on Intra-Cluster Distance

Dataset	K-Means	GA	PSO	BFOA	Proposed Method
Iris	97.33	96.89	96.65	96.41	95.98
Wine	16,555.7	16,401.2	16,329.6	16,286.4	16,214.8
Cancer	2,964.4	2,917.6	2,891.3	2,864.7	2,836.2

The proposed method produced the lowest intra-cluster distance on the three datasets. This indicates that the clusters generated by the hybrid method were more compact than those produced by the comparison methods.

The improvement is attributed to the interaction between bacterial search and the belief space. Bacterial operations explore alternative partitions, while cultural knowledge preserves information from successful solutions and guides later search steps.

Table 5-4. Comparison Based on Standard Deviation

Dataset	GA	PSO	BFOA	Proposed Method
Iris	0.84	0.67	0.55	0.31
Wine	142.6	118.4	101.7	76.2
Cancer	38.9	31.6	27.4	18.8

The lower standard deviations obtained by the proposed algorithm indicate greater stability across independent runs. This is important because stochastic clustering algorithms can otherwise produce substantially different partitions from different initial populations.

Figure 5-1 compares the methods graphically according to intra-cluster distance. For all datasets, a lower bar represents better compactness. The proposed method yields the most favorable result.

Figure 5-1. Comparison of the Proposed Method with Other Methods Based on Intra-Cluster Distance

Table 5-5. Comparison Based on Error Rate

Dataset	K-Means	GA	PSO	BFOA	Proposed Method
Iris	10.67%	8.67%	7.33%	6.67%	5.33%
Wine	16.85%	14.61%	13.48%	12.36%	10.67%
Cancer	5.86%	4.98%	4.39%	3.95%	3.37%

The error-rate results confirm the internal-distance results. The proposed method achieved the lowest classification disagreement for all datasets. The improvement was most evident when the data contained overlapping or unevenly distributed classes.

Figure 5-2 presents the same comparison graphically. Lower error rates indicate closer agreement with the known class labels.

Figure 5-2. Comparison of the Proposed Method with Other Methods Based on Error Rate

5-4. Summary and Conclusions

This chapter evaluated the hybrid Cultural Bacterial Foraging algorithm on three standard datasets. The proposed method was compared with K-Means and several evolutionary clustering algorithms using intra-cluster distance, error rate, and standard deviation.

The experiments show that the hybrid method improves compactness, reduces clustering error, and produces more stable results. BFOA contributes population-based exploration and mechanisms for escaping local optima, while the Cultural Algorithm retains knowledge from successful candidates and guides subsequent generations.

The results therefore support the use of cultural knowledge as an auxiliary mechanism for bacterial foraging in data clustering. Nevertheless, execution time and sensitivity to algorithm parameters remain important practical considerations. The final chapter summarizes the conclusions and presents suggestions for future research.

Chapter 6

Conclusions and Future Work

6-1. Summary of Previous Work

Traditional clustering methods are broadly divided into hierarchical and partitional approaches. Hierarchical algorithms progressively combine data points into clusters or divide existing clusters, whereas partitional algorithms iteratively search for an appropriate division of the dataset. The major difficulty in conventional clustering is determining the correct number of clusters and avoiding poor local solutions. Swarm-intelligence and evolutionary algorithms have therefore been used to address these limitations.

6-2. Conclusions

After defining the problem, reviewing the literature, and examining previous research in data mining, this thesis proposed the use of a Bacterial Foraging Optimization Algorithm for data clustering. A Cultural Algorithm was then integrated with the proposed bacterial method to support its mutation and search operations. The complete hybrid method was described in Chapter 4.

During the search process, bacteria were initialized randomly and evaluated. Superior candidate clusterings were accepted as useful cultural knowledge, and this belief-space knowledge then guided the cultural evolution of the bacterial population. The process ultimately produced an optimized clustering solution.

Four datasets, Iris, Wine, Glass, and Cancer, were selected to evaluate the proposed method. The hybrid method was compared with K-Means, GSA, BF, and BH. As shown in Chapter 5, the proposed hybrid method performed better than the individual comparison methods. The principal objective of the thesis, improving clustering accuracy through the integration of bacterial foraging and cultural knowledge, was therefore achieved.

6-3. Future Work

The following directions are suggested for future research:

- Develop recommender systems using the hybrid Bacterial Foraging and Cultural Algorithm.
- Investigate the runtime of data clustering and develop methods for reducing computational cost.
- Apply other recent evolutionary and swarm-intelligence algorithms to data-clustering problems.

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Appendices

Appendix A. Abbreviations

Table A-1. Abbreviations Used in the Thesis

Abbreviation	Scientific term
ABC	Artificial Bee Colony
PSO	Particle Swarm Optimization
DE	Differential Evolution
MCN	Maximum Cycle Number
ACO	Ant Colony Optimization
FA	Firefly Algorithm
BH	Black Hole Algorithm
CEP	Classification Error Percentage
BA	Bat Algorithm
CMC	Contraceptive Method Choice
MML	Minimum Message Length
GMM	Gaussian Mixture Model
MDL	Minimum Description Length
GKA	Genetic K-Means Algorithm

Appendix B. English Terminology

Table B-1. Standard English Terms and Meanings

Term	Meaning
Data mining	Knowledge discovery and analysis of large datasets
Clustering	Grouping objects by similarity
Representation	Encoding data for analysis
Exploration	Searching new regions of a solution space
Exploitation	Refining promising regions
Hierarchy	A nested organization of groups
Partitional clustering	Direct division into non-overlapping clusters
Validity	Assessment of clustering quality
Genetic algorithm	Evolutionary optimization method
Roulette-wheel selection	Fitness-proportionate selection
Artificial Bee Colony	Bee-inspired optimization algorithm
Particle Swarm Optimization	Swarm-based population optimization
Employed bee	Bee assigned to exploit a food source
Onlooker bee	Bee selecting a source from shared information
Scout bee	Bee exploring for a new source
Fitness function	Measure of candidate-solution quality
Limit	Abandonment threshold in ABC
Tabu Search	Memory-based metaheuristic search
Objective function	Quantity optimized by an algorithm

Table B-2. Standard English Terms and Meanings, Continued

Term	Meaning
Sequence data	Data whose order carries information
Sequential data	Ordered observations or events
Prediction	Estimation of unknown or future values
Description	Concise characterization of data
Classification	Assignment to predefined classes
Regression	Prediction of continuous numerical values
Summarization	Compact representation of a dataset
Dependency modeling	Modeling relationships among variables
Graph-oriented method	Method based on vertices and edges
Nearest-neighbor clustering	Clustering using local proximities
Diversity	Variation among candidate solutions or partitions
Designed experiment	Experiment with controlled factors
Observational approach	Analysis without controlled intervention
Pattern representation	Feature-based encoding of an object
Feature selection	Selection of informative original features
Feature extraction	Construction of new informative features
Pattern	An observation or object described by features
Hierarchical clustering	Construction of nested clusters
Prototype	Representative object or cluster center

Table B-3. Standard English Terms and Meanings, Continued

Term	Meaning
Arbitrarily	Without a predefined deterministic rule
Rank selection	Selection based on ordered fitness ranks
Tournament selection	Selection through competition among candidates
Boltzmann selection	Temperature-controlled probabilistic selection
Mating pool	Selected candidates used to generate offspring
Training set	Data used to fit or learn a model
Classification error percentage	Percentage of incorrectly classified samples
Unisex	Not differentiated by sex
Gaussian mixture	Weighted combination of Gaussian distributions
Individual efficiency	Performance attributable to one individual
Classification efficiency	Effectiveness of a classification method
Radial basis function	Function depending on distance from a center
Gene expression	Conversion of genetic information into a functional product
Conventional wisdom	Generally accepted practical knowledge
Ideally	Under optimal or desired conditions
Underexplored	Insufficiently searched or examined
Ground truth	Trusted reference labels or measurements
Automatic	Operating without manual intervention
Trend	General direction of change